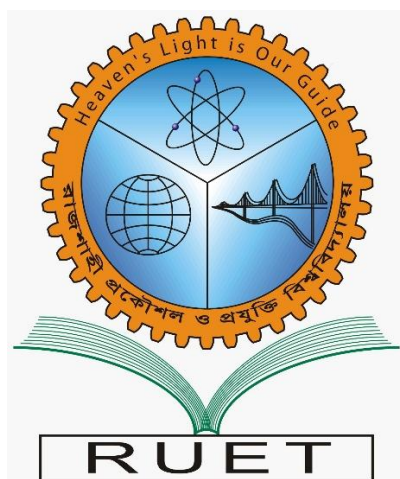


Heaven's light is our guide

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering



Sessional (Lab Report)-05

Course Name : Data Communication Sessional
Course Code : CSE 3204
Experiment No : 05
Experiment Name : QAM Modulation & Demodulation
Date of Experiment : 13 Dec, 2025
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Experiment name:**QAM Modulation and Demodulation****Objective:**

- The experiment aims to practically implement the encoding process by splitting binary data into In-phase (I) and Quadrature (Q) components and observing how they are mixed to generate the final modulated waveform.
- Furthermore, it focuses on demonstrating the demodulation process to accurately recover the original binary sequence, thereby verifying the integrity of the communication link.
- A key analytical goal is to study the Constellation Diagram (Signal Space Diagram) to visualize symbol distribution and to examine the time-domain waveforms on an oscilloscope.

Theory:

Quadrature Amplitude Modulation (QAM) is a digital modulation technique that transmits data by changing (modulating) the amplitude of two carrier waves. These two waves, usually sinusoids, are out of phase with each other by 90° and are thus called quadrature carriers. QAM is widely used because it achieves high spectral efficiency by combining both Amplitude Shift Keying (ASK) and Phase Shift Keying (PSK).

The Modulation Process:

In QAM, the input binary data stream is split into two separate streams:

- **In-phase component (I):** This stream modulates the amplitude of the cosine carrier.
- **Quadrature component (Q):** This stream modulates the amplitude of the sine carrier.

The Demodulation Process:

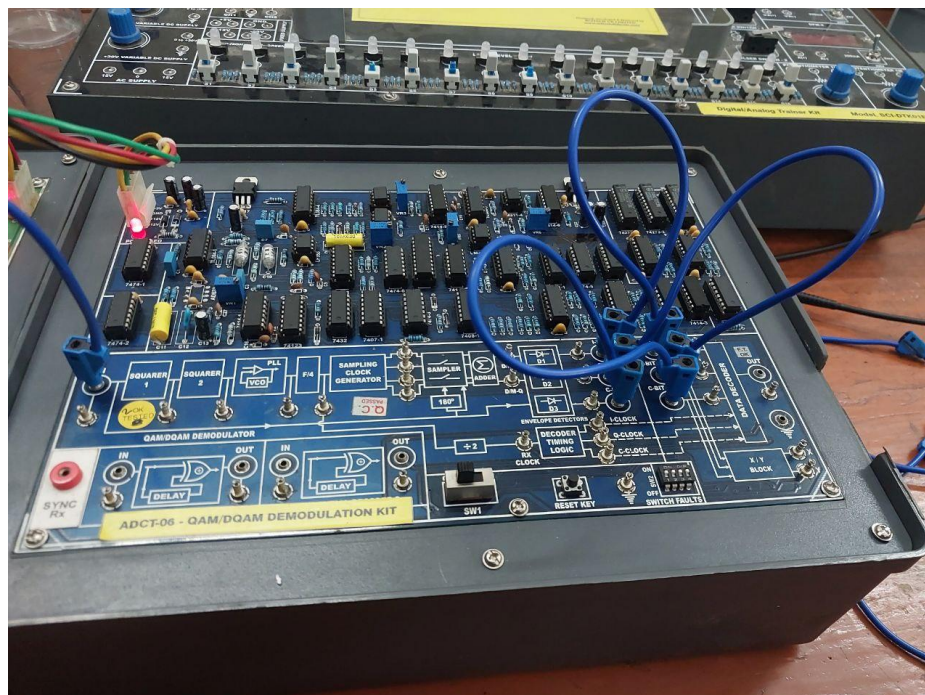
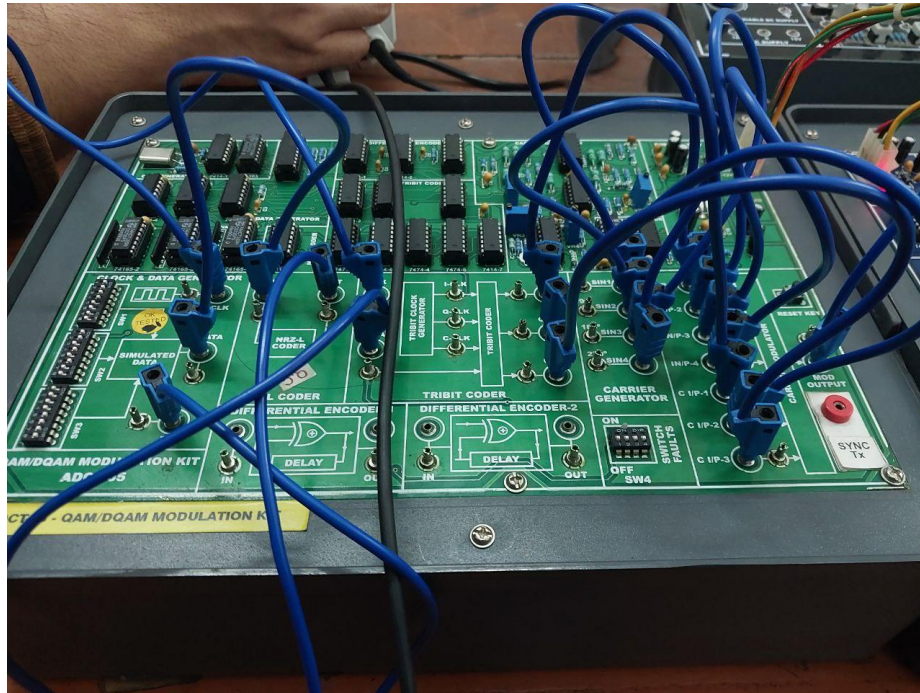
Demodulation is typically coherent, meaning the receiver must generate a local carrier that is perfectly synchronized in frequency and phase with the transmitter.

1. The received signal is split and to recover the I component and to recover the Q component.
2. Low Pass Filters (LPF) are used to remove the high-frequency double-carrier terms generated during multiplication.
3. The resulting analog voltage levels are passed through threshold detectors (ADCs) to regenerate the original binary data.

Required Equipment:

- QAM Trainer Kit
- Digital Storage Oscilloscope (DSO)
- Power Supply
- Function Generator

Pin connections:



Procedure:

QAM Modulation Process:

- **Signal Generation:** Connect the data generator to the input of the Serial-to-Parallel Converter. This splits the incoming binary stream into two parallel bits: the In-phase I bit and the Quadrature Q bit.
- **Carrier Setup:** Ensure the Sine and Cosine carrier waves are being generated at the required frequency.
- **Modulation Connection:** Connect the I-stream output to the Balanced Modulator 1 input. Connect the Q-stream output to the Balanced Modulator 2 input.
- **Summing:** Connect the output of both balanced modulators to the Adder circuit.
- **Observation:** Connect the oscilloscope probe to the output of the adder circuit. **Observe** the QAM modulated waveform. We should see amplitude and phase variations corresponding to the changing bit patterns.

QAM Demodulation Process:

- **Transmission Link:** Connect the output of the adder to the input of the QAM Demodulator.
- **Carrier Synchronization:** Connect the receiver's local carrier inputs to the same Sine and Cosine carriers used in the transmitter.
- **Filtering:** Connect the output of the first multiplier to Low Pass Filter 1. Connect the output of the second multiplier to Low Pass Filter 2.
- **Data Recovery:** Connect the outputs of the Low Pass Filters to the Comparator Circuit to regenerate the digital logic levels.
- **Parallel-to-Serial:** Connect the I and Q digital outputs to the Parallel-to-Serial Converter to merge them back into a single data stream.

Output:

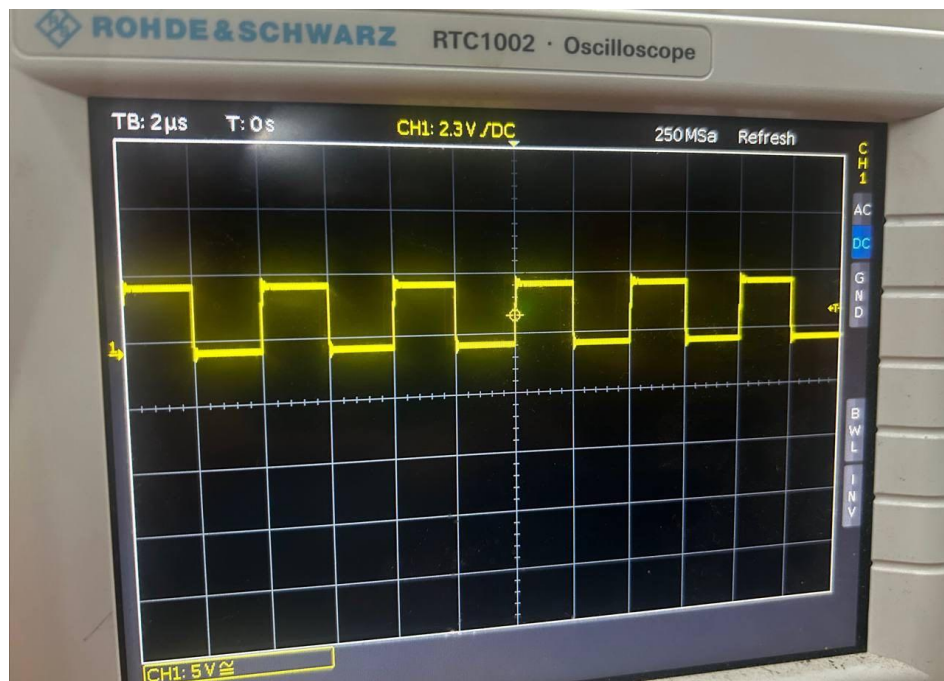


Fig-01: Input Signal

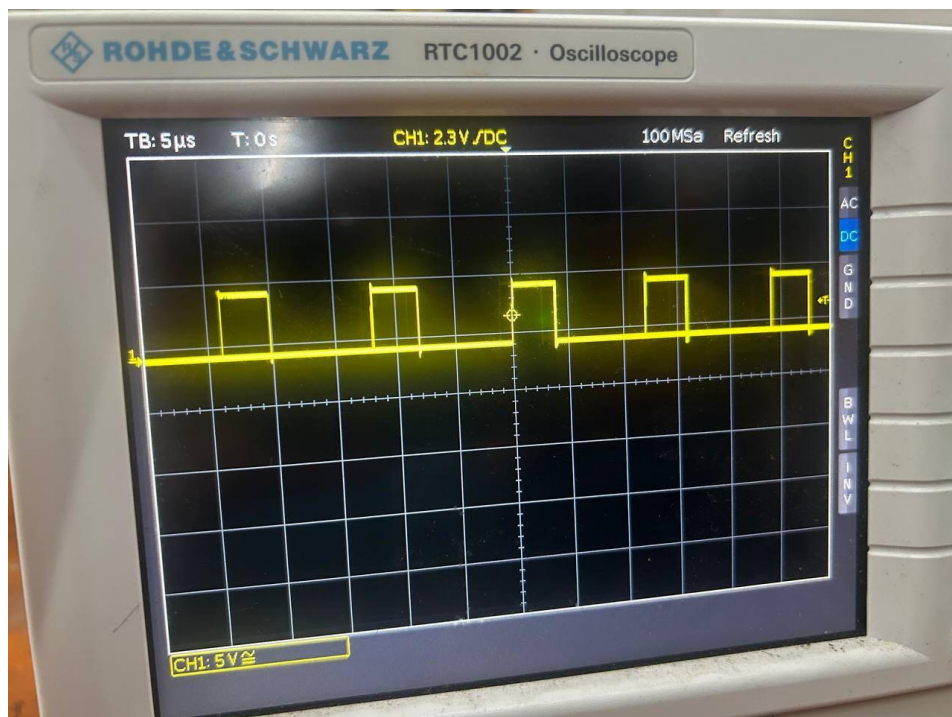


Fig-02: Output Signal

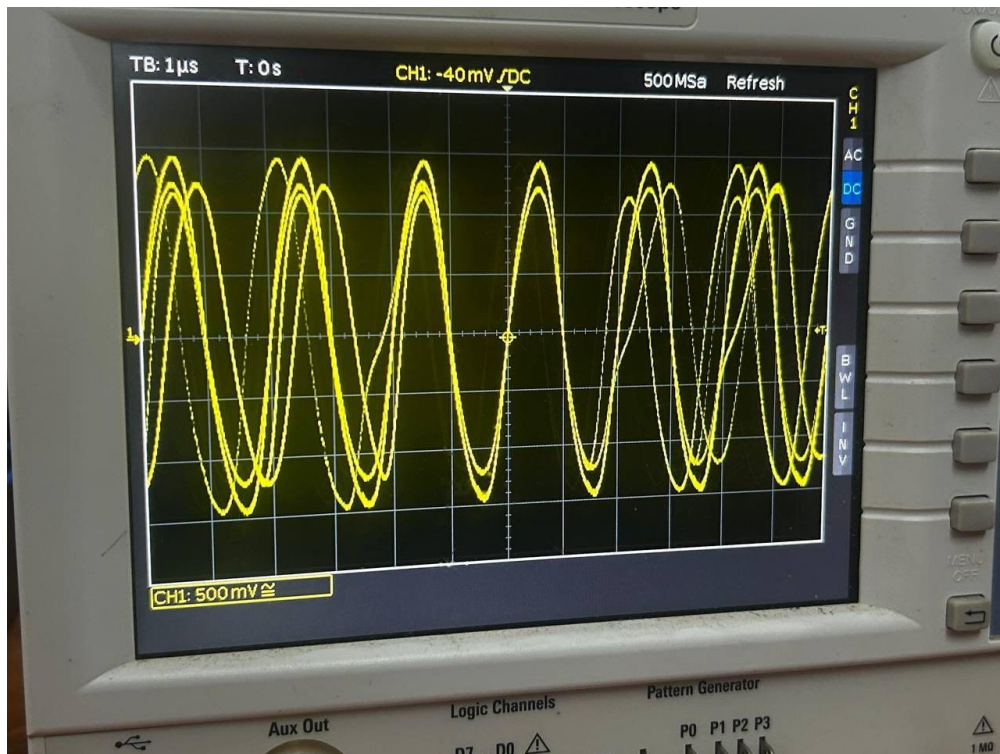


Fig-03: Modulated Signal

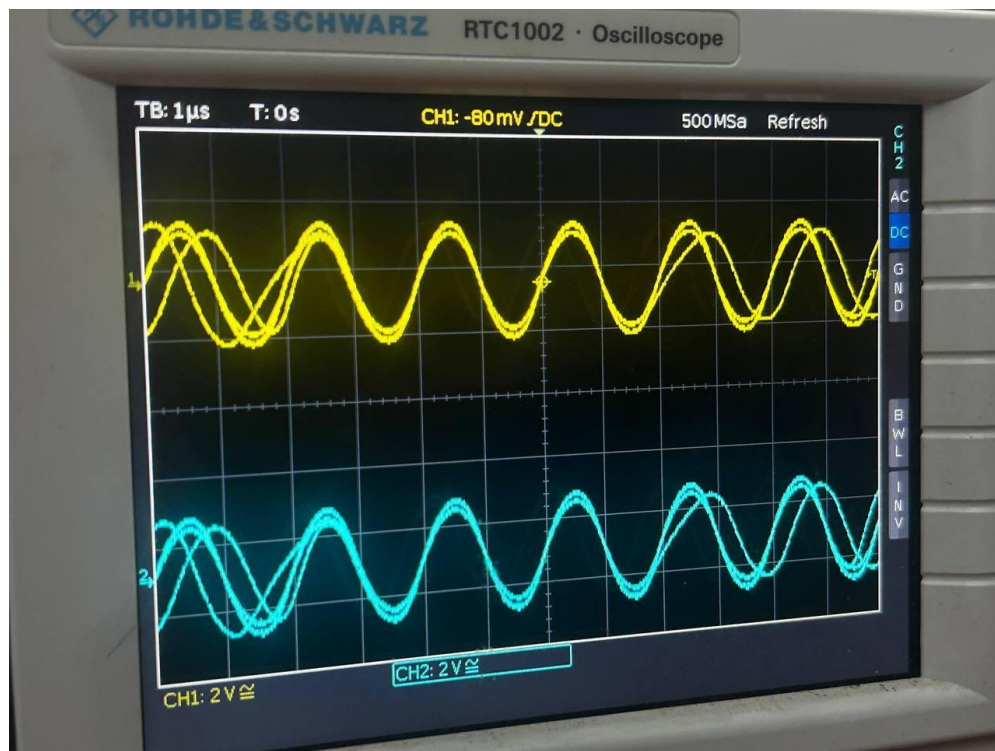


Fig-04: Modulation output and Demodulation Input

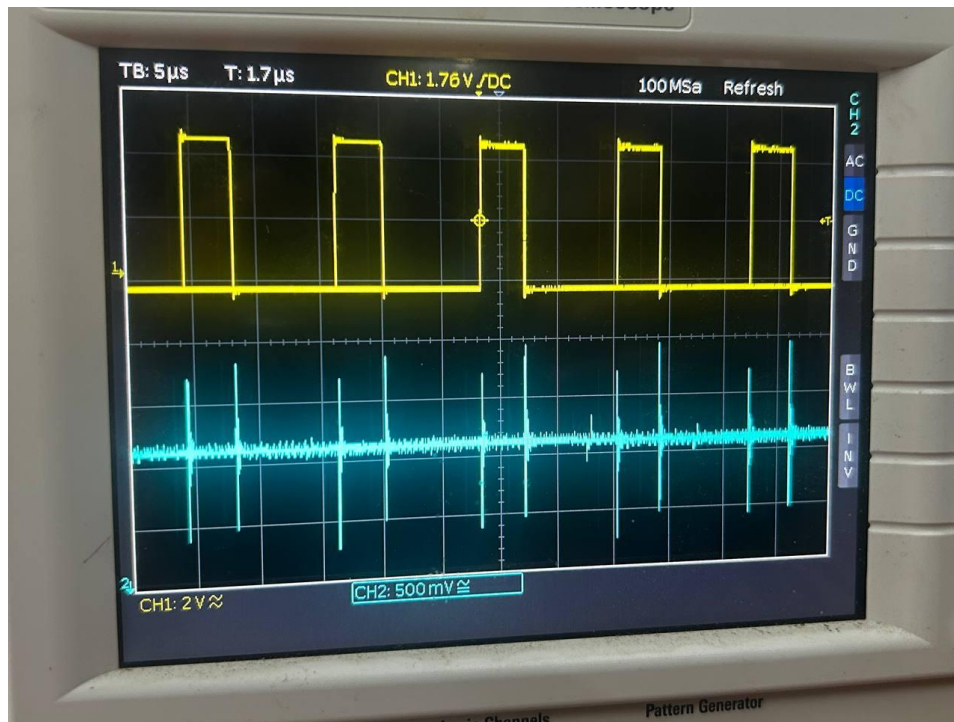


Fig-05: Output of the demodulated Signal

Precautions:

- Preventing Overmodulation
- Set Frequencies Correctly
- Properly adjust the probes of the oscilloscope

Conclusion:

In conclusion, this experiment successfully demonstrated the fundamental principles and implementation of Quadrature Amplitude Modulation (QAM), highlighting its ability to enhance spectral efficiency by modulating both the amplitude and phase of a carrier signal. By splitting the binary input into In-phase (I) and Quadrature (Q) components and mixing them with orthogonal carriers, we observed a composite waveform that carries multiple bits per symbol, offering a significant advantage over simpler schemes like ASK or BPSK. The demodulation process effectively recovered the original binary sequence through coherent detection and low-pass filtering, validating the system's accuracy and integrity. Overall, the study confirmed that while QAM significantly increases data transmission rates, it requires precise carrier synchronization and amplitude balance to maintain signal fidelity against noise and interference.